

DarkSwap Whitepaper v0.3

Version 0.3 · October 5, 2026 · Revised October 6, 2026 · Supersedes v0.2. Where this paper and the deployed contracts or live order instructions disagree, they govern.

A privacy terminal on NEAR Intents: what runs today, the dark pool on NEAR Confidential Intents that comes next, our own zero-knowledge pools after that, how \$DARK fits in, and where the limits are.

How to read this

Most sections carry a status tag. The tag tells you how much weight to put on what follows.

LIVE

Running today with real funds.

BUILT

Code exists and runs end to end on local chains. Not on any public network.

SCHEDULED

Designed, with an owner and a target date.
Can still change.

PROPOSED

Decided in direction, not yet built or dated.
Likely to change in detail.

Still to decide. These are open and the paper does not assume an answer.

Who owns the liquidity that ZK pool fees add to the ZEC-DARK pool: a permanently locked position, or a DarkSwap-held position.

Terms of DarkSwap's NEAR Intents partner agreement, including fees for the dark pool on NEAR.

Multisig signers, the Solana upgrade authority and the audit firm for the ZK tier.

01

LIVE

Summary

DarkSwap is a privacy terminal built on NEAR Intents. It does three things, at three levels of maturity.

- **Private swaps, live.** Swaps from Solana to other chains with confidential routing. No account, and nothing moves until you send the deposit yourself from your own wallet.
- **The dark pool on NEAR, next.** A hidden balance you can deposit into from a supported chain, hold, swap inside and withdraw from, built on NEAR Confidential Intents. Target: late October 2026, subject to NEAR enabling access for DarkSwap. It is a target, not a commitment.
- **Our own ZK pools, later.** Zero-knowledge shielded pools for Solana, Ethereum and Base. Built and tested on local chains, shown on the site as a local-development preview, and held back until public testnets, a production setup ceremony and an independent audit.

\$DARK, the project token, is paired with NEAR and pays its holders in wNEAR from the creator fee on \$DARK trades. A ZEC-DARK liquidity pool is live on Meteora, and fees from our own ZK pools are to be added to it.

This paper separates what runs today from what is planned. Every section carries a status tag; read the tag before the text.

02

The problem

A wallet is a public record. Every balance and every payment on Solana and EVM chains is open to anyone, permanently. One labelled address, from an exchange withdrawal, a payment or a public post, can tie the rest of a wallet's history to a person.

Today's fixes each leave a gap.

OPTION	GAP
Mixers	A secret to keep for every deposit, and a public link at both ends.
Zcash	The longest-running shielded design, but it needs its own wallet and an exit back to where the liquidity is.
A one-off private swap	Hides the processing; the deposit and the destination still sit side by side.

DarkSwap's answer is a hidden balance you can hold, swap and withdraw from, on a network that already reaches the chains people use, with the trust model stated plainly at each tier.

03

LIVE

SCHEDULED

BUILT

Architecture

DarkSwap is a terminal, not a custodian. It prepares quotes and orders, shows the user exactly what to send, and tracks the result. Execution happens on infrastructure DarkSwap does not run.

LAYER	WHAT IT DOES	STATUS
Terminal (darkswap.app)	Quotes, order review, deposit instructions, order tracking, the pool interface	LIVE
NEAR Intents 1Click	Cross-chain routing and settlement for Privacy swap, with confidential handling requested	LIVE
Partner route	A second private route through a partner provider	LIVE
NEAR Confidential Intents	Confidential balances on NEAR's private shard for the dark pool	SCHEDULED
DarkSwap ZK pools	Shielded note pools on Solana, Ethereum and Base	BUILT

NEAR says Intents coordinates execution across more than 35 chains. Which assets can be swapped varies by route; a supported network does not mean every asset pair has a live route.

04

LIVE

Private swaps today

Two private routes from Solana to other chains are open on darkswap.app.

01

Quote

Choose what you send and where it should land. The terminal requests a live quote, including every fee.

02

Review

Check the estimate, the fee and the destination address before anything happens. Creating an order moves no funds.

03

Deposit yourself

Send the exact amount from your own wallet to the order's deposit address.
DarkSwap never signs a transaction for you and never holds your funds.

04

Track

Follow the order until it settles or is refunded by the provider.

Privacy swap runs on the NEAR Intents 1Click API and requests confidential handling for the order. NEAR's documentation says that for Confidential Intents, on-chain deposits and withdrawals cannot be tracked to each other. Your Solana deposit is still public: its amount and the wallet it came from.

Since October 4, 2026, Privacy swap quotes include a 0.40% partner fee, charged from the input and shown in every quote. Under NEAR's standard partner schedule, 0.20% goes to DarkSwap and 0.20% to NEAR Intents 1Click. It is earned only on completed swaps.

Private route runs through a second partner provider, with its own fees and terms shown in the quote.

A private route keeps more of the swap off the public record. It does not make a swap anonymous, and timing and amounts at either end can still be matched.

05

SCHEDULED

The dark pool on NEAR

This is the intended design, not a live product. It depends on NEAR enabling Confidential Intents for DarkSwap (access is invite-only and is being arranged with the NEAR Intents team) and on DarkSwap completing the integration.

Confidential Intents is a private transaction layer on NEAR Intents. It already powers Confidential Mode on near.com and, according to Infinex's launch report, confidential accounts on Infinex since July 2026. Balances live on a NEAR private shard, a separate execution environment run by permissioned validators and connected to NEAR mainnet by a bridge. The private shard has no public RPC, API or block explorer.

- 01 Deposit from a supported chain**
Send SOL, ETH, USDC or another supported asset from the wallet you already use. The deposit is public, and NEAR Intents screens funds as they enter.
- 02 Hold in the dark**
The funds become a confidential balance on NEAR Intents. The public cannot see the balance or what you do with it.
- 03 Swap privately**
Swaps from the confidential balance execute on the private shard. Amounts, routes and pairs are not recorded on the public chain.
- 04 Withdraw to a supported chain**
The withdrawal is public. NEAR's documentation says it cannot be tracked back to your deposit on-chain; a matching amount or timing can still give the link away.

Swaps and withdrawals from a confidential balance are authorised by signed intents: the terminal asks NEAR Intents to generate the exact intent, your own wallet signs it, and the signed intent is submitted. DarkSwap does not hold your keys or your balance; the balance lives on NEAR Intents.

NEAR describes this as encryption and restricted validator access rather than zero-knowledge proofs: no proof is generated on your device, and the trust model is trust-minimised rather than trustless. NEAR also plans user-held view keys for selective disclosure, and states that it discloses to law enforcement only under valid legal process.

NEAR Intents offers no test environment for this integration, so the beta runs on mainnet with small amounts of DarkSwap's own funds first.

06

BUILT

Our own ZK pools

The ZK pools are not deployed to any public network and have not been audited. Their proving keys are development keys: whoever ran that setup could forge proofs. Do not put real funds in them.

DarkSwap has written its own shielded pools from one circuit: a contract for Ethereum and Base (ETH and listed ERC-20 tokens) and a native Solana program (SOL and listed SPL tokens). Both run end to end on local chains with real proofs. The terminal for them is on the site at darkswap.app/pool as a local-development preview, not deployed on any public network. This is the ZK tier, not the Dark Pool on NEAR Confidential Intents.

Balances are encrypted notes, in the model Zcash uses. A note commits to an amount, an asset, an owner key and a random blinding value. Spending a note publishes a nullifier that marks it spent without saying which note it was.

- **Shield.** A public deposit creates a note. Each deposit uses a fresh one-time key, so deposits from one wallet cannot be linked to each other on-chain.
- **Private send.** Spends up to two notes and creates two, proven with a zero-knowledge proof built on your device in 2 to 5 seconds. A private send hides the amount and the asset as well as who sent it.
- **Unshield.** A withdrawal to a public address, submitted by a relayer so you do not need gas in the destination wallet. The relayer fee is separate from the protocol fee.

Both pools share a depth-26 Poseidon note tree, per-asset books (one asset can never pay out another's funds), deposit minimums, maximums and caps per asset. The pool moves one asset per transaction; there are no swaps inside it yet.

Admin powers. The admin can list assets, set limits, turn deposits off and pause new deposits until a set date. It cannot move user funds, pause withdrawals or change the verifying key. Withdrawals can still be affected by the asset itself (for example, a token whose issuer can freeze accounts), by the network, or by the deployment.

Upgrades. The Ethereum and Base pool has no upgrade path. The Solana program has an upgrade authority, as every Solana program does; before mainnet it goes behind a multisig with a delay, or is removed.

Review. Two separate reviews attacked the code, one on the circuit and one on both pools. Neither found a way to steal, mint, double-spend or redirect a payout. The pool review found two ways to freeze funds; both are fixed and each fix has a test. These reviews are not an independent audit.

Gates before user funds. Public testnets, a setup ceremony with outside contributors to replace the development proving keys, and an independent audit with its findings resolved.

07

LIVE

BUILT

Fees and revenue

Four revenue lines feed the project: three are running today and one is built into the ZK design.

LINE	WHERE IT COMES FROM	WHERE IT GOES	STATUS
Creator fee	1% on \$DARK trades, received in wNEAR	Half to holder payouts in wNEAR; half buys ZEC tokens on Solana for the treasury in the creator wallet	LIVE
Swap fee	0.40% partner fee on Privacy swap; 0.20% to DarkSwap	DarkSwap's NEAR account	LIVE

LINE	WHERE IT COMES FROM	WHERE IT GOES	STATUS
Liquidity fees	DarkSwap's own positions in the ZEC-DARK pool earn their share of trading fees	A share planned for operating costs and, later, swap subsidies; not yet funded	LIVE
ZK pool fee	0.5% on shield and on unshield; none on private sends; capped at 1% in the contract	Converted to ZEC and \$DARK and added to ZEC-DARK liquidity	BUILT

Pool fees compound the ZEC-DARK pool. ZK pool fees never sit inside a note balance and never make an existing note worth more. They are swept to one fee recipient fixed when the pool is deployed. That recipient is designed to convert them to ZEC and \$DARK and add them as liquidity to the ZEC-DARK pool on Meteora, with fees from Ethereum and Base bridged to Solana first. Anyone can trigger each step; no step can send the fees anywhere else. The recipient contracts are not written yet and will be audited with the pool.

Provider and network costs come on top of DarkSwap's fees and are set by those providers. Fees and rewards exist only when people swap and trade \$DARK.

Holding \$DARK does not create a claim on any of these fees.

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LIVE

\$DARK holder rewards

\$DARK is paired with NEAR. Half of received \$DARK creator fees funds qualifying holders in wNEAR. The other half purchases ZEC tokens on Solana for the protocol treasury in the creator wallet. Treasury holdings, separately authorized liquidity deployments, LP fee compounding, and completed holder payouts are tracked separately. The program never shares keys or balances with the pools.

- **What is split.** Only creator-fee proceeds actually received, in wNEAR. Not all trading volume, router commissions or customer deposits. There is no buyback and burn of

creator fees, and buying \$DARK to pair in a liquidity pool is not burning it.

- **Scheduled wNEAR payouts.** At 6 a.m. and 6 p.m. America/Los_Angeles time, adjusting for daylight saving, sent direct with no claim step. Each payout uses a fresh snapshot taken at that payout. Background monitoring snapshots do not set the payout's weight and are not a countdown to an automatic ZEC payout.
- **How to qualify.** Hold at least 100,000 \$DARK in the scheduled snapshot. Excluded wallets remain ineligible. Allocations are weighted by qualifying balances in that snapshot. Earlier awards keep the rules they were paid under.
- **ZEC award, historical.** A completed loyalty award distributed 1 ZEC token on Solana across 92 holders on October 5, 2026, weighted by holdings under its original checkpoint rules. Further ZEC awards require separate approval.
- **Which ZEC.** ZEC here is a token on Solana, not native or shielded Zcash. A matching ticker does not establish backing, redemption rights or shielding.

Payout totals are not restated here. The DARK Rewards console at rewards.darkswap.app is the record, with a summary feed and a receipt export. It reports holder payouts, \$DARK marketing airdrops, treasury assets and pool liquidity separately, and its USD totals are current-price estimates, not payout-time value or guaranteed proceeds. Rewards depend on fees actually received. There is no fixed return, and past distributions do not guarantee future ones.

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LIVE

PROPOSED

ZEC-DARK liquidity

A protocol-funded ZEC-DARK pool is live on Meteora (Solana, DAMM v2). It has a 1% base trading fee with dynamic fees where applicable, so the fee paid can be higher, and 10% of LP fees are configured to compound back into the pool. That 10% is a compounding setting, not a 10% yield.

The protocol reports that the \$DARK creator wallet funded 99.9% of the ZEC-DARK pool’s liquidity. Reported initial funding was 2 ZEC used to buy \$DARK, paired with another 2 ZEC. These figures are protocol-reported until the purchase and deposit transactions are linked. Funding is separate from position ownership and withdrawable liquidity: the creator-held position includes vested and permanently locked portions. About 2.212 ZEC was observed in the pool on October 5, 2026; that is a dated observation, not a current balance, and balance growth alone does not prove compounded fee earnings.

It is public trading liquidity. It is not the dark pool, it is not shielded, and its figures are separate from both the dark pool and the holder rewards.

Two inflows are planned on top of trading: treasury ZEC from the creator-fee split, only when a liquidity deployment is separately authorized, and fees from DarkSwap's own ZK pools once they are live (section 07). DarkSwap also plans to set aside a share of the LP fees it actually claims for operating costs first and then a small, capped swap-subsidy trial. None of those reserves is funded yet.

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Who you trust

DarkSwap has two privacy tiers with two trust models.

	DARK POOL ON NEAR	DARKSWAP ZK POOLS
Privacy from	The public. Not from the operators of NEAR's private shard	Everyone, by proof, for what happens inside the pool
How	Encrypted execution on a private shard with permissioned validators	Encrypted notes and zero-knowledge proofs built on your device
Who holds funds	NEAR Intents; you authorise every action with your own wallet	The pool contract; only the note owner can spend

	DARK POOL ON NEAR	DARKSWAP ZK POOLS
Screening	NEAR Intents screens funds as they enter, under its own rules	Asset listing and deposit limits set by the admin
Can withdrawals be stopped?	Subject to NEAR Intents' availability and terms	No role can pause withdrawals; the asset, network or deployment still can affect them

Other parties on both tiers: the routing providers that carry funds between chains, whose availability and terms are outside DarkSwap's control, and the chains themselves.

On both tiers, privacy from the public is not privacy from a lawful request.

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What privacy you get, and what you do not

Every tier hides activity in the middle. None hides the edges. Someone watching the chain can see that you deposited an amount and that someone withdrew an amount; the question is whether they can connect the two.

- **Amount and timing.** An unusual amount withdrawn soon after a matching deposit can be linked, on any tier. Common amounts and waiting help.
- **Small pools hide less.** In the ZK pools, a small pool is a small crowd to hide in. NEAR's confidential balances are encrypted per user rather than pooled, but edge correlation still applies.
- **Your own habits.** Reusing addresses, posting transaction details or withdrawing to a labelled wallet undoes privacy no system can restore.

No route or pool makes activity invisible. This paper does not claim otherwise, and neither should anyone describing it.

How it compares

The two peers reviewed here each run their pool on one chain. DarkSwap builds on a network that already connects many chains, and arrives with a live product and paid holders.

	ZEROTRACE	NULLMASK	DARKSWAP
Chains	Robinhood Chain	EVM; pool on Ethereum	Chains supported by NEAR Intents (routes vary); ZK tier built for Solana, Ethereum and Base
Privacy model	Zero-knowledge notes	Zero-knowledge notes, through a wallet proxy	Confidential routing live; confidential balances planned; zero-knowledge notes later
Assets	ZERO only	Multi-asset	Assets supported by NEAR Intents, where a route is available
Holder payouts	Through the vault, to private notes	None stated	wNEAR paid direct to holders; one ZEC-token award paid
Live product	Deployed	Deployed; about \$13.1k TVL on Oct 2, 2026	Private swaps live; dark pool planned
Audit of own contracts	No public report identified	No public report identified	None yet; required before the ZK tier takes funds

Peer columns are project-reported, as of October 4, 2026. “No public report identified” means none was found in this review; it does not mean none exists.

Roadmap

STAGE	WHAT
Live	Private swaps from Solana through NEAR Intents (with a partner fee) and a partner route. wNEAR payouts and one ZEC-token award to \$DARK holders. The ZEC-DARK pool on Meteora.
On the site	The ZK pool terminal at darkswap.app/pool, a local-development preview, not deployed on any public network.
Next	The dark pool on NEAR Confidential Intents. Target late October 2026, subject to NEAR enabling access; beta on mainnet with DarkSwap's own funds first.
Built, held back	Our own ZK pools for Solana, Ethereum and Base, then the fee recipient contracts that add pool fees to ZEC-DARK liquidity. Public testnets, a setup ceremony and an independent audit come before any user funds.

Targets move; the gates do not.

Risks

- **The dark pool depends on NEAR.** Confidential Intents is invite-only and DarkSwap's access is not confirmed yet. NEAR can change its terms, fees or availability.
- **Confidential is not zero-knowledge.** The NEAR tier hides activity from the public, not from the operators of NEAR's private shard.
- **No test environment for NEAR Intents.** Testing happens on mainnet with small amounts of DarkSwap's own funds.

- **The ZK tier is unaudited and not deployed.** It has run on local chains only, with development proving keys.
- **Routes vary.** Not every asset pair has a live route, and providers can pause routes.
- **Revenue depends on use.** Fees, rewards and liquidity inflows exist only when people swap and trade \$DARK.
- **Market risk.** \$DARK, wNEAR and ZEC prices move; reward values and pool liquidity move with them.
- **Legal.** No legal opinion has been obtained. Privacy tools draw regulatory attention.

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Key numbers

The figures in this paper, with their status. A built or proposed number is a design value, not a commitment.

ITEM	VALUE	STATUS
Privacy swap partner fee	0.40% of the input; 0.20% to DarkSwap, 0.20% to NEAR Intents 1Click (standard partner schedule)	LIVE
\$DARK creator fee	1% on \$DARK trades, received in wNEAR; half to holder payouts in wNEAR, half buys ZEC tokens on Solana for the treasury	LIVE
Holder reward minimum	100,000 \$DARK in a fresh snapshot at each 6 a.m. / 6 p.m. Pacific payout	LIVE
ZEC-DARK pool fee	1% base fee (dynamic fees enabled); 10% of LP fees configured to compound into the pool	LIVE

ITEM	VALUE	STATUS
ZK pool protocol fee	0.5% on shield and on unshield, none on private sends; contract cap 1%	BUILT
ZK pool fee destination	Converted to ZEC and \$DARK and added to ZEC-DARK liquidity	PROPOSED
ZK note tree	Poseidon, depth 26; last 1,000 roots accepted on EVM, last 256 on Solana	BUILT
ZK transaction shape	Up to 2 notes in, 2 out, one asset per transaction	BUILT
Proving time	2 to 5 seconds per transaction on the user's device	BUILT
Dark pool on NEAR target	Late October 2026, subject to NEAR enabling access	SCHEDULED

This paper explains; it does not promise.

It is a summary for information only. It is not an offer to sell tokens or securities, and it is not investment advice. DarkSwap is built on NEAR Intents and is not endorsed by NEAR Foundation.